

POCKET SMART AI

Your Smart Budget & Recommendation Assistant

PROJECT SUBMISSION DOCUMENT

Technology Track: Generative AI with Google

COLLEGE / INSTITUTION LOGO

INSERT ACTUAL PROJECT EVIDENCE HERE
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INSERT ACTUAL PROJECT EVIDENCE HERE

Student Name	_____
Register / Roll No.	_____
Department / Class	_____
Institution	_____
Guide / Mentor	_____
Academic Year	_____

Reference: the supplied PDF identifies PocketSmart AI as Project No. 1 under “Generative AI with Google” and provides a project document link and reference video link.

1. REFERENCE PROJECT

The supplied PDF lists the project as follows:

Item	Reference
Technology Track	Generative AI with Google
Project	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Project Document	Google Docs project document link supplied in the reference PDF
Reference Video	YouTube reference video link supplied in the reference PDF

The supplied PDF is a one-page project list. It provides the project title and links, but not the detailed eight-phase implementation. The detailed requirements in this document are therefore a structured implementation plan prepared for submission.

2. PHASE 1 — IDEATION & PROBLEM IDENTIFICATION

Problem Statement

Users may find it difficult to consistently track expenses, understand category-wise spending, maintain a monthly budget, and convert raw transaction data into practical insights. Traditional expense trackers can show numbers but may not explain patterns in natural language or provide conversational assistance.

Proposed Solution

USER → Income & Expenses → Database → Analysis → Budget Engine → Generative AI → Recommendations → USER

Objectives

- Record and organize income and expenses.
- Categorize transactions and provide searchable transaction history.
- Allow users to create and monitor budgets.
- Track savings goals and progress.
- Visualize spending and income through charts.
- Provide a conversational Generative AI assistant.
- Generate personalized, data-grounded budgeting recommendations.
- Produce reports and a clear project demonstration.

Evidence to Collect

PHOTO 1 — TEAM BRAINSTORMING

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Figure 1: Team discussion and brainstorming for Pocket Smart AI.

PHOTO 2 — PROJECT REFERENCE / DISCUSSION

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INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Figure 2: Analysis of the reference project and project idea.

3. PHASE 2 — REQUIREMENTS & PLANNING

Functional Requirements

Module	Requirements
User Management	Registration, login, logout, profile management
Income	Add, edit, delete, view income
Expenses	Add, edit, delete, categorize, search, filter
Budget	Create, edit, monitor, compare spending with budget
Savings Goals	Create target, set date, track progress
AI Assistant	Ask questions, analyze spending, generate recommendations
Reports	Summaries, charts, export-ready report views
Notifications	Budget threshold / overspending alerts

Non-Functional Requirements

- Security: password hashing, protected sessions, secure API-key storage, input validation.
- Performance: responsive database operations and reasonable AI response time.
- Usability: simple navigation, readable dashboards, responsive interface.
- Reliability: accurate calculations, persistent storage, error handling.
- Maintainability: separate modules for authentication, database, AI, budgeting and UI.

Hardware Requirements

Minimum	Recommended
4 GB RAM	8 GB RAM or more
10 GB free storage	20 GB+ free storage
Modern browser	Modern browser
Internet connection	Stable internet connection

Software Requirements

Component	Recommended Technology
Operating System	Windows / Linux / macOS
IDE	Visual Studio Code
Backend	Python + Flask
Database	SQLite
Frontend	HTML5 + CSS3 + JavaScript
UI	Bootstrap
Charts	Chart.js
AI	Google Generative AI / Gemini
Version Control	Git / GitHub

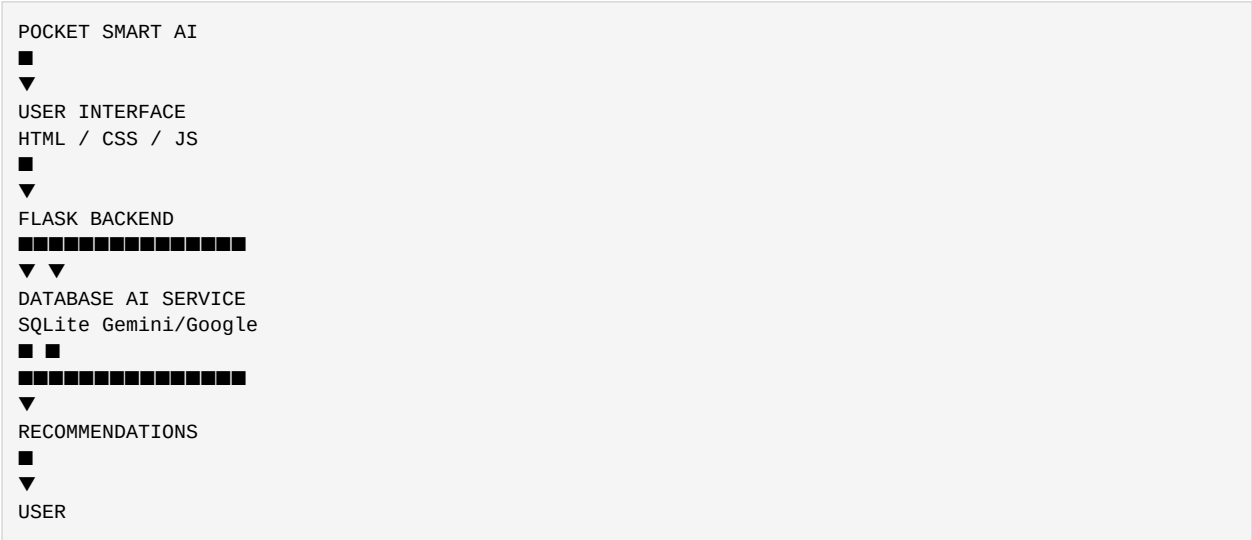
PHOTO 3 — REQUIREMENTS / PLANNING

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Figure 3: Requirements analysis and project planning.

4. PHASE 3 — UI/UX & SYSTEM DESIGN

System Architecture



Main UI Screens

SCREENSHOT 1 — LOGIN

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INSERT ACTUAL PROJECT EVIDENCE HERE

Figure: Login screen with email, password and registration link.

SCREENSHOT 2 — REGISTRATION

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INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Figure: Registration form.

SCREENSHOT 3 — DASHBOARD

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INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Figure: Income, expenses, balance, savings and charts.

SCREENSHOT 4 — ADD EXPENSE

INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Figure: Amount, category, description and date.

SCREENSHOT 5 — BUDGET

INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Figure: Category budgets and utilization.

SCREENSHOT 6 — SAVINGS GOALS

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INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Figure: Target, saved amount and progress.

SCREENSHOT 7 — AI ASSISTANT

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INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Figure: Conversational AI interface.

5. DATABASE & DESIGN SPECIFICATION

Users Table

Field	Type	Purpose
user_id	INTEGER	Primary key
name	TEXT	User name
email	TEXT	Login email
password	TEXT	Hashed password
monthly_income	REAL	Income reference
created_at	DATETIME	Account creation time

Transactions Table

Field	Type	Purpose
transaction_id	INTEGER	Primary key
user_id	INTEGER	Owner
type	TEXT	Income / Expense
amount	REAL	Transaction amount
category	TEXT	Expense category
description	TEXT	Description
date	DATE	Transaction date

Budgets, Goals and AI Recommendations

Table	Key Fields
budgets	budget_id, user_id, category, amount, month
goals	goal_id, user_id, goal_name, target_amount, current_amount, target_date
ai_recommendations	recommendation_id, user_id, recommendation, created_at

Diagrams to Include

- Use Case Diagram
- Data Flow Diagram (DFD)
- Entity Relationship Diagram (ERD)
- Activity Diagram
- Sequence Diagram
- Application Flowchart

DIAGRAM — USE CASE / DFD / ERD

INSERT ACTUAL PROJECT EVIDENCE HERE
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Insert final diagrams generated for the implemented project.

6. PHASE 4 — APPLICATION DEVELOPMENT

Recommended Project Structure

```
PocketSmartAI/  
■■■ app.py  
■■■ config.py  
■■■ requirements.txt  
■■■ .env  
■■■ database/  
■ ■■■ pocketsmart.db  
■■■ models/  
■■■ services/  
■■■ templates/  
■■■ static/
```

Core Development Modules

- Authentication module — registration, login, logout and protected sessions.
- Transaction module — income/expense CRUD operations.
- Budget module — budget creation and monitoring.
- Goal module — savings goal creation and progress.
- Dashboard module — totals, charts and recent transactions.
- AI service — prompt construction, API request and response handling.
- Recommendation service — combines deterministic calculations with AI explanations.

Development Screenshots

SCREENSHOT 8 — REGISTRATION PAGE

INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Insert actual application screenshot here.

SCREENSHOT 9 — LOGIN PAGE

INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Insert actual application screenshot here.

SCREENSHOT 10 — DASHBOARD

INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Insert actual application screenshot here.

SCREENSHOT 11 — ADD EXPENSE

INSERT ACTUAL PROJECT EVIDENCE HERE
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INSERT ACTUAL PROJECT EVIDENCE HERE

Insert actual application screenshot here.

SCREENSHOT 12 — TRANSACTION HISTORY

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INSERT ACTUAL PROJECT EVIDENCE HERE

Insert actual application screenshot here.

SCREENSHOT 13 — BUDGET PAGE

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INSERT ACTUAL PROJECT EVIDENCE HERE

Insert actual application screenshot here.

SCREENSHOT 14 — SAVINGS GOAL

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INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Insert actual application screenshot here.

SCREENSHOT 15 — CHARTS

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Insert actual application screenshot here.

7. PHASE 5 — GENERATIVE AI INTEGRATION

AI Role

The Generative AI component should turn structured financial information into understandable natural-language responses. The application should ground prompts in the user's actual transaction, budget and goal data and avoid inventing transactions.

AI Features

- Conversational financial assistant.
- Spending analysis.
- Personalized budgeting recommendations.
- Savings suggestions.
- Natural-language explanation of budget status.
- AI-assisted expense categorization.
- Goal-based savings explanation.
- Budget generation based on user-provided information.

Example Prompt Template

You are Pocket Smart AI, a personal budgeting assistant.

Monthly Income: {income}

Expenses: {expenses}

Budget: {budget}

Savings Goal: {goal}

Provide:

1. Spending summary
2. Budget observations
3. Potential savings areas
4. Practical recommendations

Use clear language and only use the supplied data.

AI Demonstration Screenshots

SCREENSHOT 16 — AI ASSISTANT OPEN

INSERT ACTUAL PROJECT EVIDENCE HERE
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INSERT ACTUAL PROJECT EVIDENCE HERE

Figure: AI assistant interface.

SCREENSHOT 17 — AI SPENDING ANALYSIS

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INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Figure: User asks the AI to analyze spending.

SCREENSHOT 18 — AI RECOMMENDATION

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INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Figure: AI provides a recommendation.

SCREENSHOT 19 — AI BUDGET REQUEST

INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Figure: User asks for a budget.

SCREENSHOT 20 — AI RESPONSE

INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Figure: AI-generated budget explanation.

AI Safety Note: display that the application provides informational budgeting assistance and does not replace professional financial advice. Keep API keys server-side and never commit them to source control.

8. PHASE 6 — TESTING & VALIDATION

ID	Test Case	Input / Action	Expected Result	Actual
T01	Registration	Valid details	Account created	_____
T02	Login	Valid credentials	Dashboard opens	_____
T03	Login error	Wrong password	Error shown	_____
T04	Add expense	■500 Food	Expense saved	_____
T05	Delete expense	Existing record	Record deleted	_____
T06	Budget	Create category budget	Budget stored	_____
T07	Overspending	Expense > budget	Alert shown	_____
T08	AI query	Ask spending question	AI response	_____
T09	Savings goal	Create target	Goal displayed	_____
T10	Charts	Transactions exist	Chart displayed	_____

Error Testing

- Empty fields.
- Invalid email.
- Negative or invalid amount.
- Duplicate account.
- Invalid date.
- AI API unavailable.
- Internet disconnected.
- Database error.

Testing Evidence

SCREENSHOT 21 — INVALID LOGIN

INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Insert actual test screenshot.

SCREENSHOT 22 — VALIDATION / ERROR

INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Insert actual test screenshot.

SCREENSHOT 23 — BUDGET ALERT

INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Insert actual test screenshot.

9. PHASE 7 — DEPLOYMENT & DEMONSTRATION

Demonstration Flow

```
graph TD; A[Open Application] --> B[Login]; B --> C[Dashboard]; C --> D[Add Income]; D --> E[Add Expenses]; E --> F[Set Budget]; F --> G[Create Savings Goal]; G --> H[View Charts]; H --> I[Ask AI]; I --> J[Receive Recommendation]; J --> K[Show Final Dashboard];
```

Screen Recording Plan

Time	Content
0:00–0:30	Title and project introduction
0:30–1:00	Problem and proposed solution
1:00–1:30	Registration / login
1:30–2:30	Dashboard and transactions
2:30–3:30	Budget management
3:30–4:15	Savings goals
4:15–6:30	Generative AI demonstration
6:30–7:00	Charts / reports
7:00–7:30	Conclusion

Screen Recording Checklist

- Application opening
- Login
- Dashboard
- Income entry
- Expense entry
- Budget
- Savings goal
- Charts
- AI chatbot
- AI recommendation
- Final result

SCREEN RECORDING COVER / VIDEO THUMBNAIL

INSERT ACTUAL PROJECT EVIDENCE HERE
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Insert a screenshot from the final recording. Submit the actual video separately.

10. PHASE 8 — DOCUMENTATION & FINAL PRESENTATION

Final Report Structure

1. Title Page
2. Certificate
3. Declaration
4. Acknowledgement
5. Abstract
6. Introduction
7. Problem Statement
8. Objectives
9. Existing System
10. Proposed System
11. Requirements
12. System Design
13. Database Design
14. Implementation
15. Generative AI Integration
16. Testing
17. Results
18. Screenshots
19. Conclusion
20. Future Enhancements
21. References

PPT Presentation — Recommended 15–20 Slides

Slide	Topic
1	Title
2	Introduction
3	Problem Statement
4	Objectives
5	Existing System
6	Proposed System
7	Technology Stack
8	Architecture
9	Modules
10	Database
11	Generative AI

Slide	Topic
12	UI Screenshots
13	AI Demonstration
14	Testing
15	Results
16	Advantages
17	Limitations
18	Future Enhancements
19	Conclusion
20	Thank You / Q&A

11. PHOTO, SCREENSHOT & VIDEO MASTER CHECKLIST

Replace every placeholder in this document with actual evidence from the completed application before final academic submission.

Evidence	Required	Collected
Team brainstorming photo	Yes	■
Requirements/planning photo	Yes	■
UI/design photo	Yes	■
Coding/development photo	Yes	■
Database photo	Yes	■
AI integration photo	Yes	■
Testing photo	Yes	■
Final demonstration photo	Yes	■
Login screenshot	Yes	■
Registration screenshot	Yes	■
Dashboard screenshot	Yes	■
Expense screenshots	Yes	■
Budget screenshot	Yes	■
Goal screenshot	Yes	■
Chart screenshots	Yes	■
AI assistant screenshot	Yes	■
AI recommendation screenshot	Yes	■
Testing/error screenshots	Yes	■
Screen recording	Yes	■
PPT presentation	Yes	■
Final project report	Yes	■

Recommended screen-recording duration: approximately 5–8 minutes. Demonstrate the working application rather than spending most of the recording on source code.

12. ADVANTAGES, LIMITATIONS & FUTURE ENHANCEMENTS

Advantages

- Combines conventional budgeting with Generative AI assistance.
- Provides conversational explanations instead of only numerical dashboards.
- Organizes income, expenses, budgets and savings goals in one interface.
- Visualizes spending patterns.
- Modular design allows future expansion.

Limitations

- AI responses depend on the quality and completeness of supplied data.
- AI API availability and usage limits may affect responses.
- The prototype is not a banking or accounting system.
- Recommendations are informational and not professional financial advice.

Future Enhancements

- Mobile application.
- Voice-based assistant.
- Receipt image recognition.
- Multi-language support.
- Advanced trend analysis.
- Secure external financial-data integrations where appropriate.
- More detailed reporting and export options.

13. CONCLUSION

Pocket Smart AI is designed as a Generative AI-based budgeting and recommendation assistant that brings together transaction tracking, budgeting, savings goals, data visualization and conversational AI. The eight-phase plan provides a complete path from ideation through requirements, design, implementation, AI integration, testing, demonstration and final documentation.

For the final submission, replace the marked placeholders with actual application screenshots, project-development photographs, diagrams and a screenshot from the completed screen recording. The screen recording itself should be submitted as a separate video file.

FINAL APPLICATION SCREENSHOT

INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE
INSERT ACTUAL PROJECT EVIDENCE HERE

Insert final working application screenshot.

14. REFERENCES

1. Supplied reference PDF: Project Details - Generative AI. It lists PocketSmart AI as Project No. 1 under Generative AI with Google and provides the associated project-document and reference-video links.
2. Google Generative AI / Gemini official documentation applicable to the chosen API/SDK.
3. Flask official documentation applicable to the implementation.
4. SQLite official documentation applicable to the database.
5. Chart.js official documentation applicable to dashboard charts.

SUBMISSION NOTE

This PDF is a complete project-submission specification and evidence template. It does not fabricate screenshots or photographs that were not supplied. Insert the actual evidence from your completed Pocket Smart AI application before submitting it for evaluation.